

SeasonalTemp: Temperature Forecasting Based on Time Series

1. Introduction

Project goal: Forecasting the maximum temperature based on data.

Key tasks:

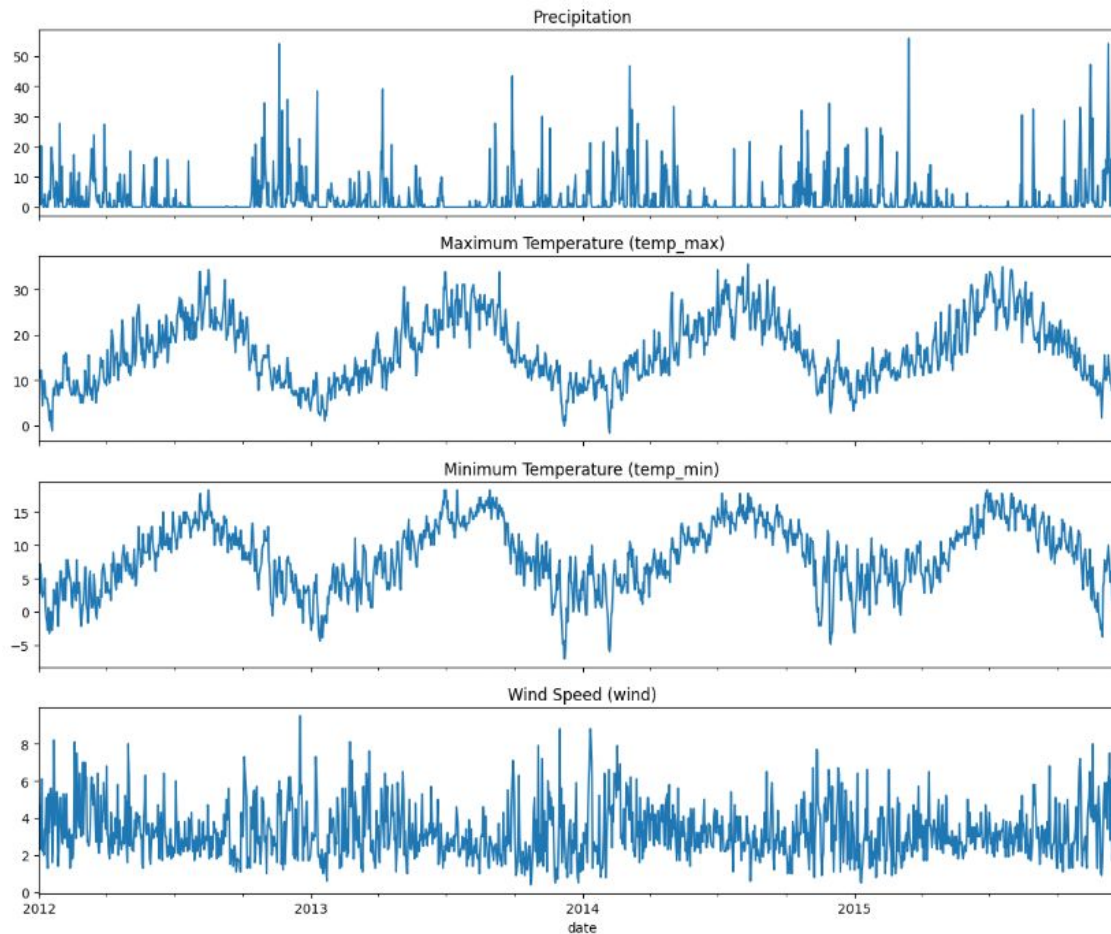
- Exploratory Data Analysis (EDA)
- Data cleaning and preparation
- Model selection
- Forecast evaluation

2.Exploratory Data Analysis (EDA)

Before modeling, the following data analysis was performed:

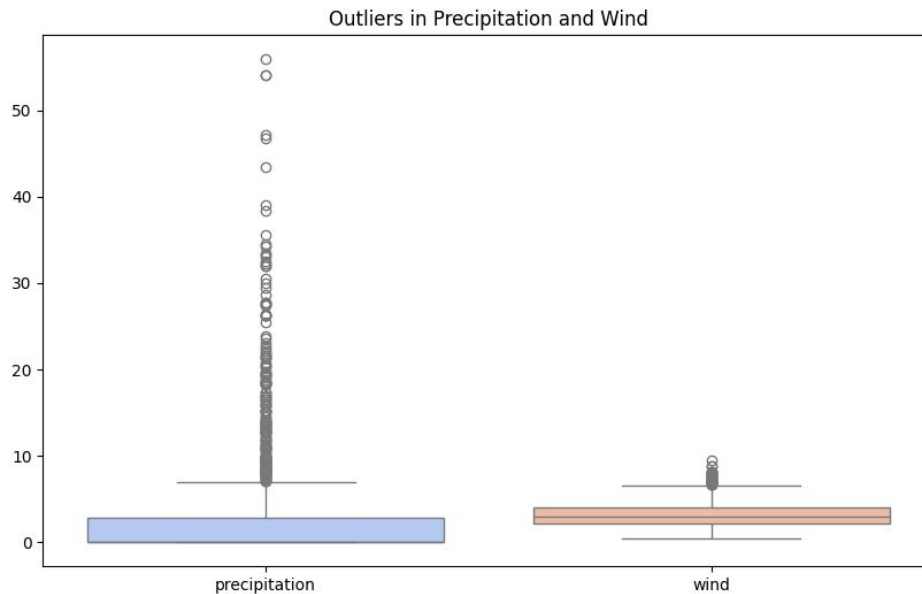
- Time series visualization
- Outlier detection
- Correlation analysis
- Seasonality and trend analysis

2.1 Time Series Visualization



- A clear seasonality was identified.
- There is high variability in precipitation and wind speed.

2.2 Outlier Analysis

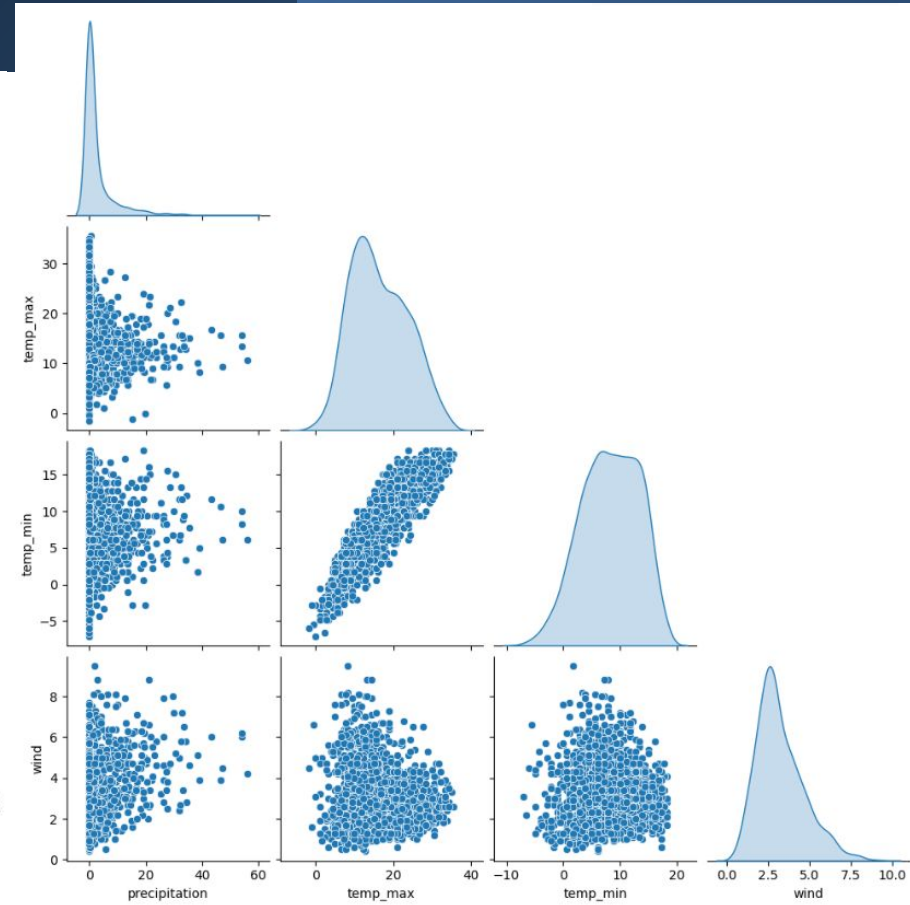
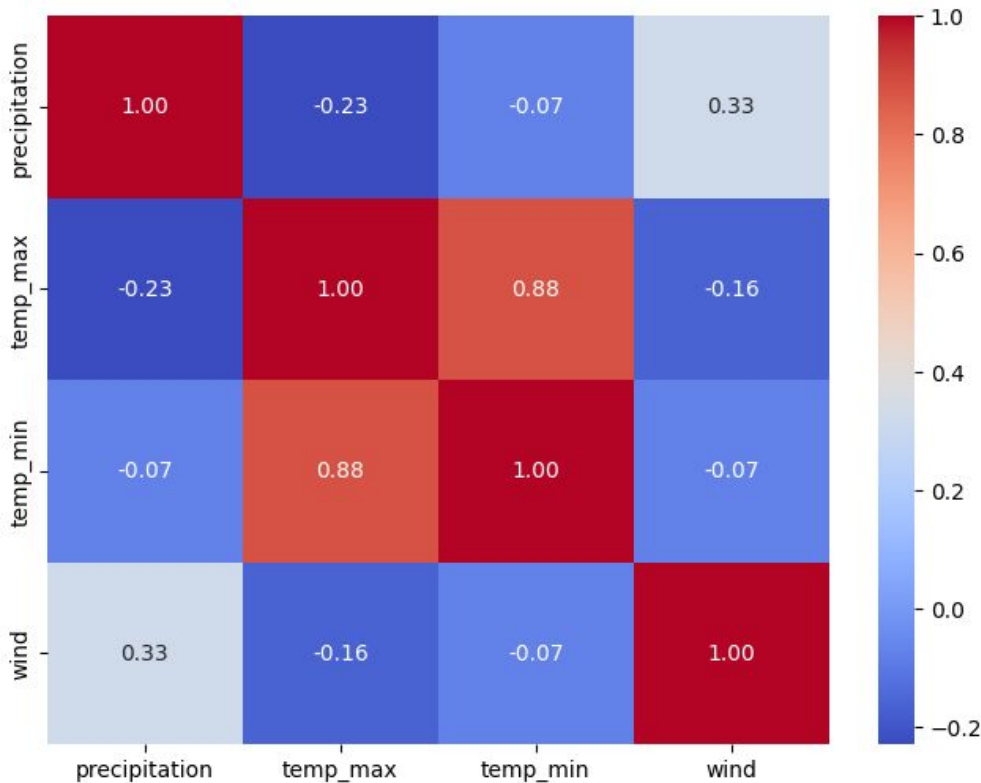


Number of outliers based on Z-score:

- Precipitation: 37
- Max Temperature: 0
- Min Temperature: 1
- Wind: 15

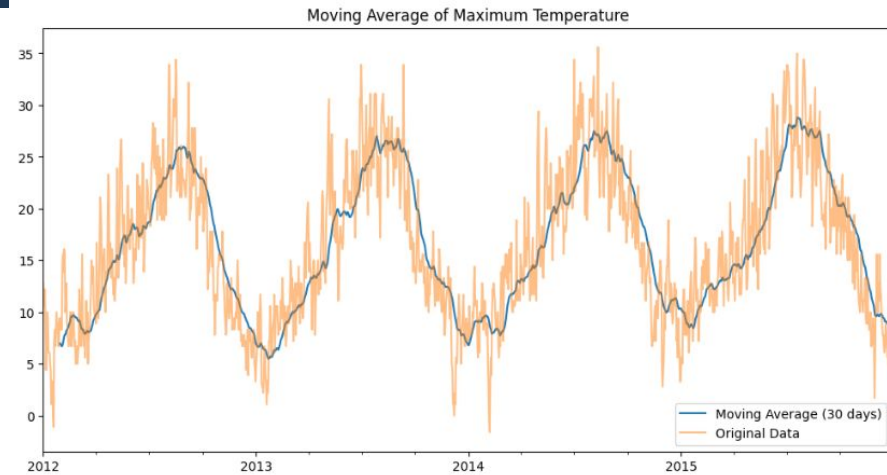
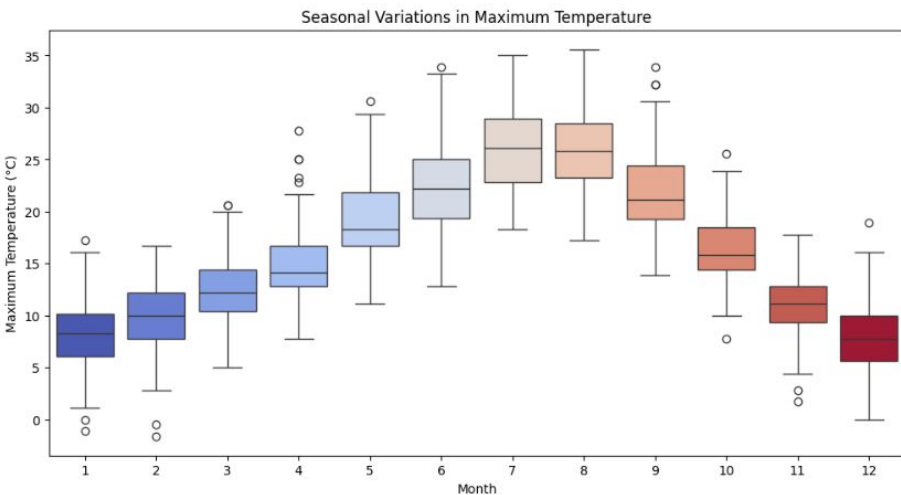
Outliers were found in precipitation and wind speed.

2.3 Correlation Analysis

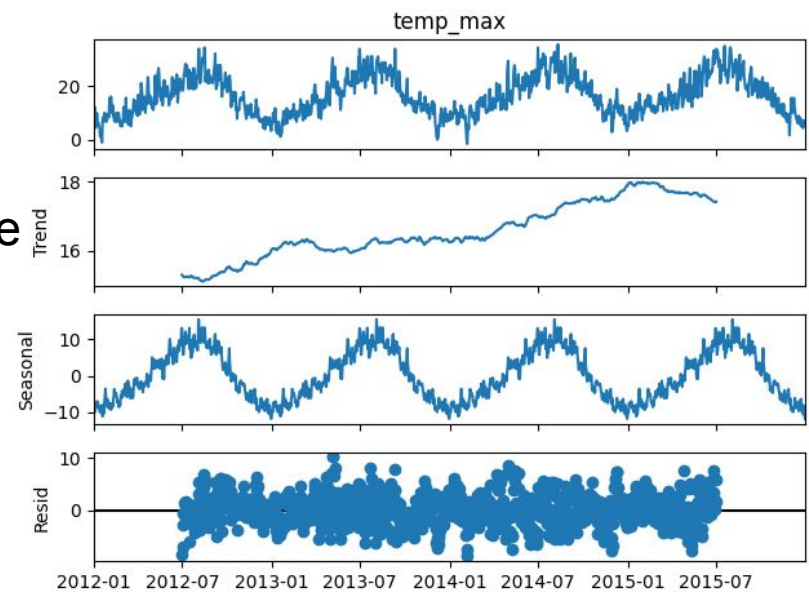


- Max temperature is strongly correlated with min temperature.
- Precipitation and wind have a weak relationship with temperature.
- These variables have been included as regressors.

2.4 Seasonality and Trend Analysis



- The moving average plot (30 days) confirms the annual seasonality.
- The boxplot by month shows summer and winter fluctuations.
- Seasonality is considered in the models.



Stationarity of the Time Series

Augmented Dickey-Fuller (ADF) Test Results (Before Differencing)

Variable	ADF Statistic	p-value	Stationarity
Precipitation	-16.80	0.0000	✓ Yes
temp_max	-2.43	0.1337	✗ No
temp_min	-2.61	0.0918	✗ No
Wind Speed	-24.51	0.0000	✓ Yes

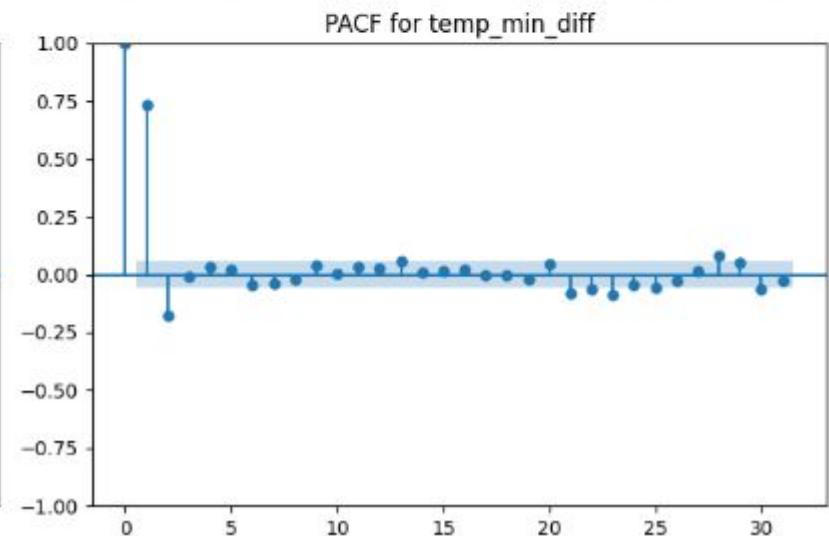
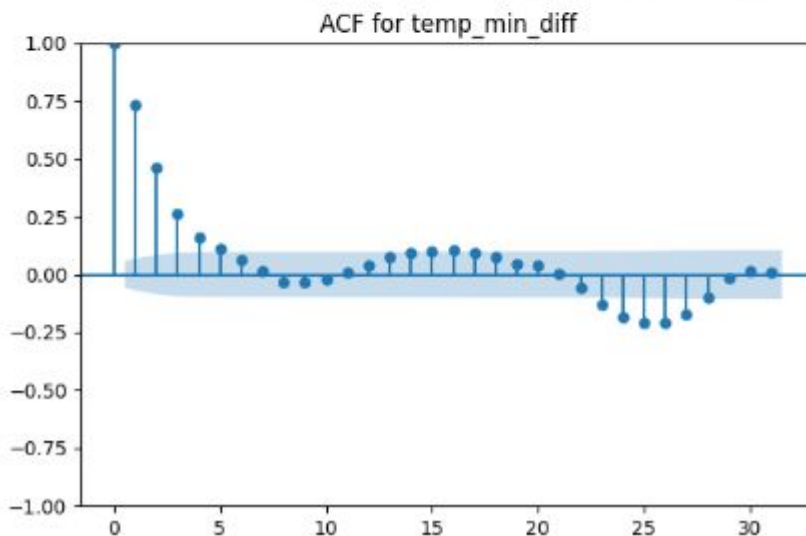
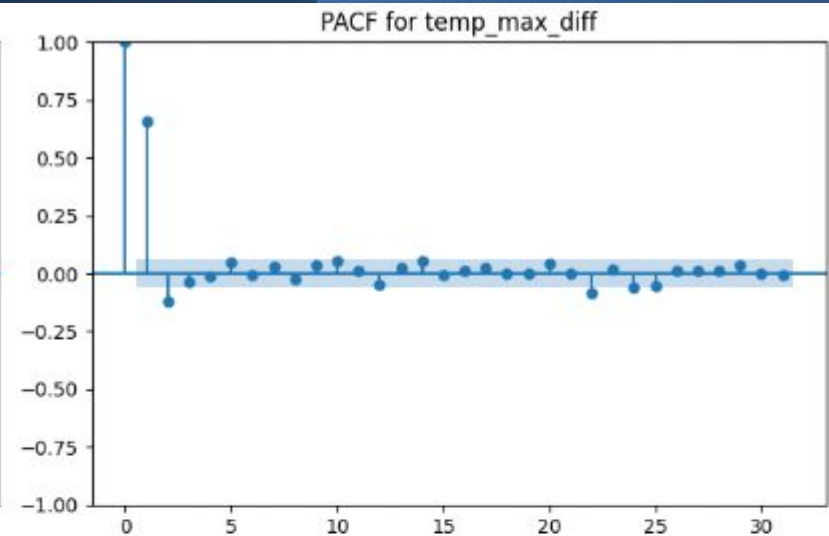
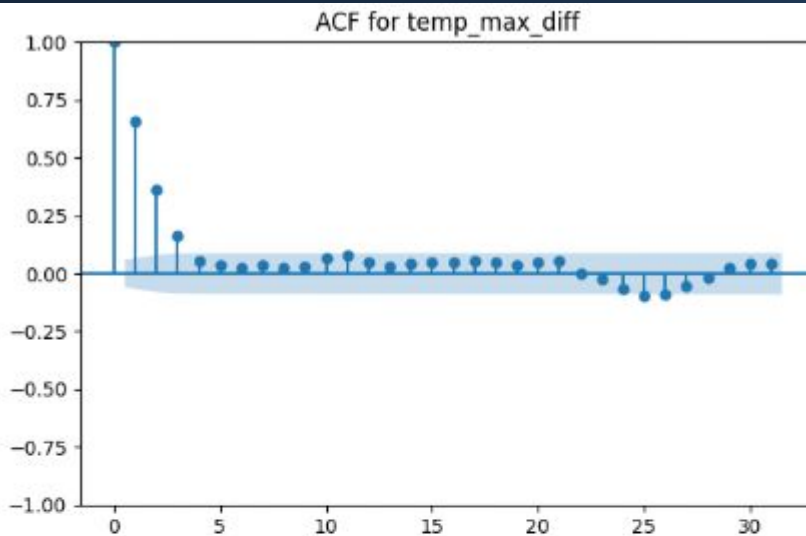
Augmented Dickey-Fuller (ADF) Test Results (After Seasonal Differencing)

Variable (after differencing)	ADF Statistic	p-value	Stationarity
temp_max_diff	-15.55	0.0000	✓ Yes
temp_min_diff	-14.41	0.0000	✓ Yes

3. Model Selection and Their Parameters

Model	Parameters
SARIMA	ARIMA (1,1,1) $\times$ (0,1,1,54) – Seasonal model
Prophet + Reg	Multiplicative seasonality, external regressors
Prophet_Corrected_SARIMA	Prophet prediction + SARIMA residual correction

ACF/PACF to Determine Parameters p , q , P , Q



3.1 SARIMA (Seasonal ARIMA)

SARIMA (Seasonal AutoRegressive Integrated Moving Average) is an extended version of ARIMA that accounts for seasonality in the data.

General formula for SARIMA(p,d,q)(P,D,Q,s):

$$Y_t = \mu + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t$$

$$(1 - \phi_1 B)(1 - B^D)Y_t = (1 + \theta_1 B)(1 + \Theta_1 B^s)\epsilon_t$$

- p – order of autoregression (AR)
- d – degree of differencing (I)
- q – order of moving average (MA)
- P, D, Q – seasonal counterparts of ARIMA parameters
- s – length of the seasonal period

3.2 Prophet + Enhanced Regressors

Prophet is a time series model developed by Facebook that automatically identifies trends, seasonality, and other patterns.

The Prophet model formula: $Y_t = g(t) + s(t) + h(t) + \epsilon_t$

- $g(t)$ – trend component (piecewise linear/logistic)
- $s(t)$ – seasonality (annual, monthly, weekly)
- $h(t)$ – influence of regressors (external variables)
- ϵ_t – model errors

Added regressors:

- Min temperature
- Precipitation
- Wind speed
- Day of the week, month, lags

3.3 Prophet_Corrected_SARIMA

Comparison of Prophet and SARIMA Models

Criterion	SARIMA	Prophet
Seasonality	Requires manual tuning (parameters P, D, Q, s)	Automatically detected
Trend	Modeled through differencing	Linear or logistic trend
Regressors	Not supported	Can include external variables (precipitation, wind, etc.)
Handling Outliers	Sensitive to outliers	Robust to anomalies
Automation	Requires parameter selection (ACF/PACF)	Almost fully automated
Autocorrelation Handling	Effectively corrects autocorrelation	May still have autocorrelation

This model combines the capabilities of Prophet and SARIMA:

- Prophet makes the primary prediction – it accounts for trends, seasonality, and regressors.
- SARIMA corrects the residuals of Prophet – it removes autocorrelation in the errors.

Combined approach:

- Prophet makes the primary prediction.
- SARIMA corrects the autocorrelated residuals.
- Improves the structure of temporal errors.

Prophet_Corrected_SARIMA is the most accurate model as it combines the strengths of both methods.

4. Model Quality Evaluation

- Mean Absolute Error (MAE) – the mean absolute error.

→ The lower the value, the more accurate the forecast.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- Root Mean Squared Error (RMSE) – the root of the mean squared error.

→ It shows how much, on average, the forecast deviates from the actual values.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Ljung-Box Test – testing residuals for white noise.

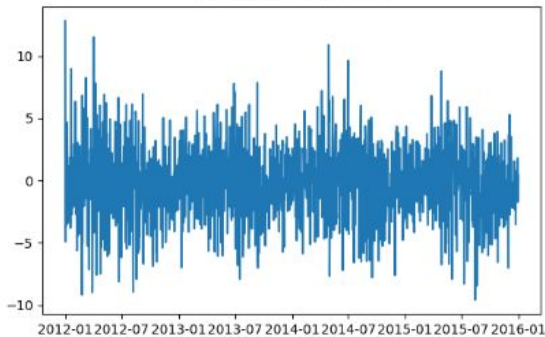
→ If the p-value > 0.05, the residuals are random, indicating a well-calibrated model.

Model	MAE ↓	RMSE ↓	Ljung-Box (p-value)
SARIMA	2.83	3.59	0.9673 
Prophet + Enhanced Regressors	1.33	2.03	0.9729 
Prophet_Corrected_SARIMA	1.30	1.98	0.9673 

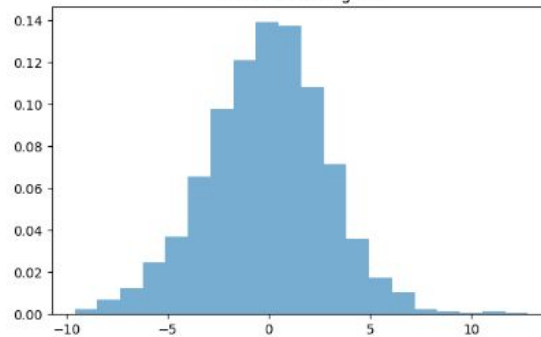
Model Diagnostics

SARIMA

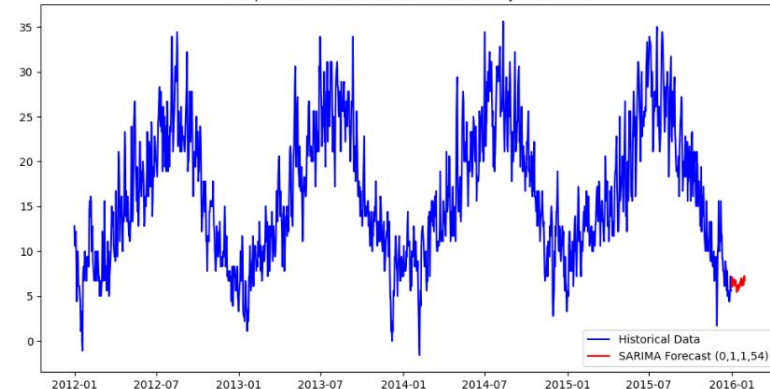
Standardized Residuals



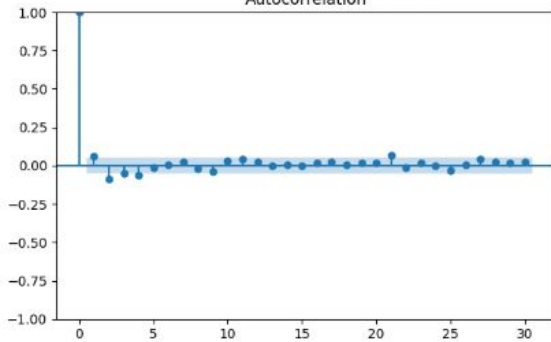
Residuals Histogram



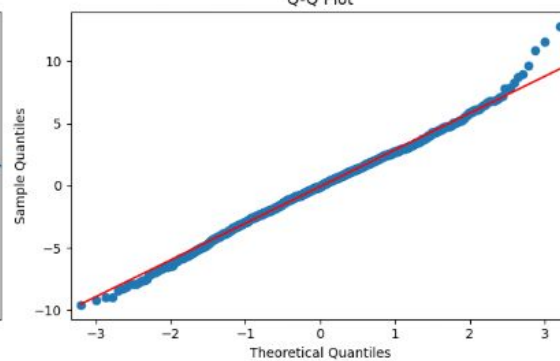
Temperature Forecast (SARIMA, Seasonality 54, No AR)



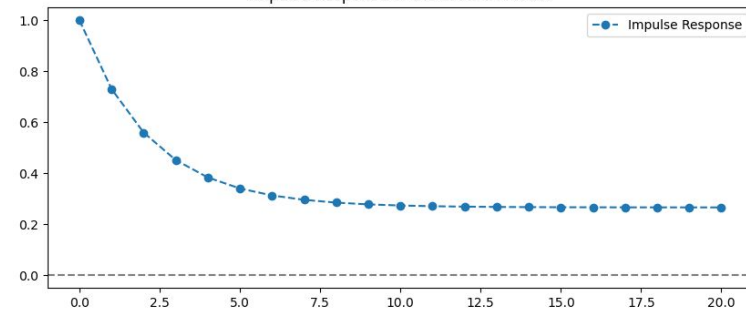
Autocorrelation



Q-Q Plot

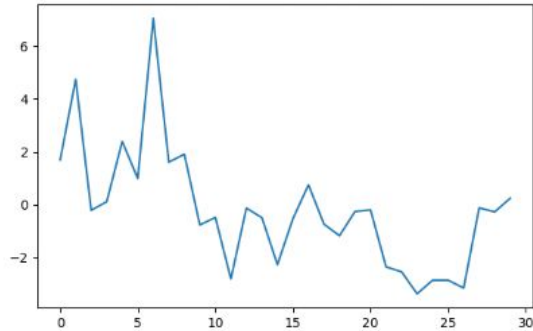


Impulse Response of the SARIMA Model

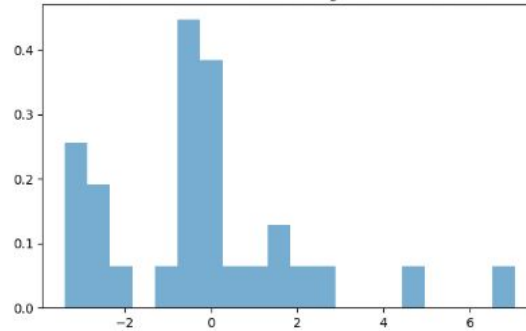


Prophet + Enhanced Regressors

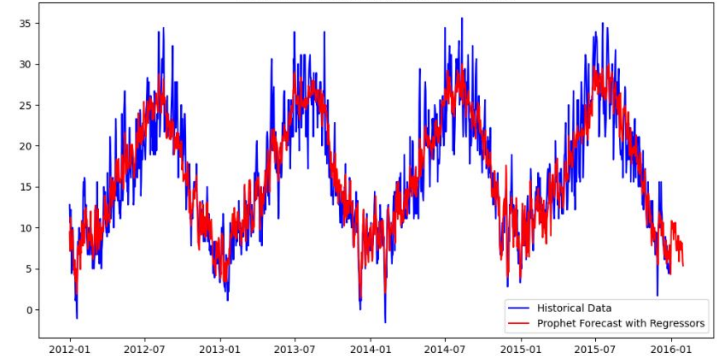
Standardized Residuals



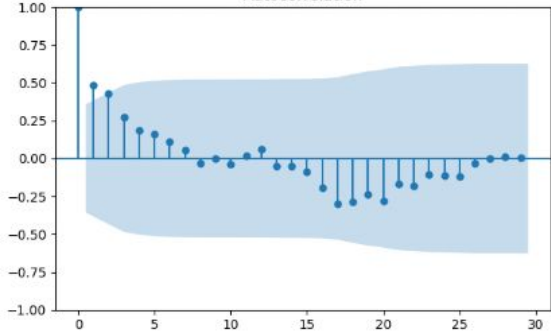
Residuals Histogram



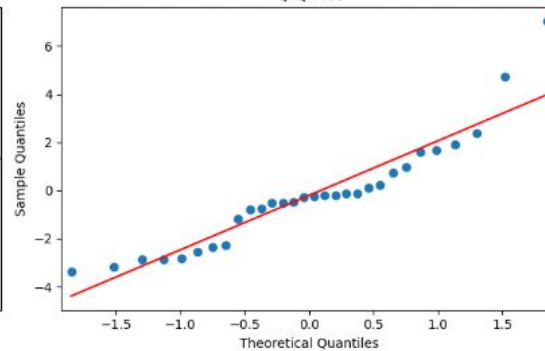
Temperature Forecast (Prophet + Regressors)



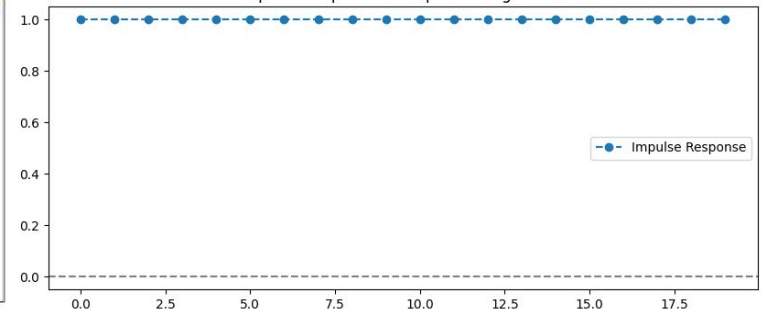
Autocorrelation



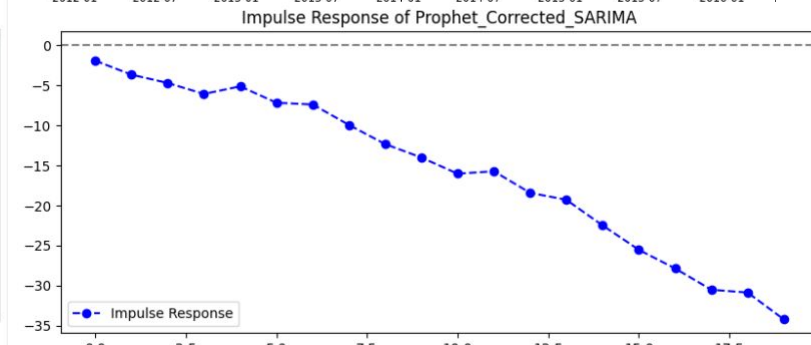
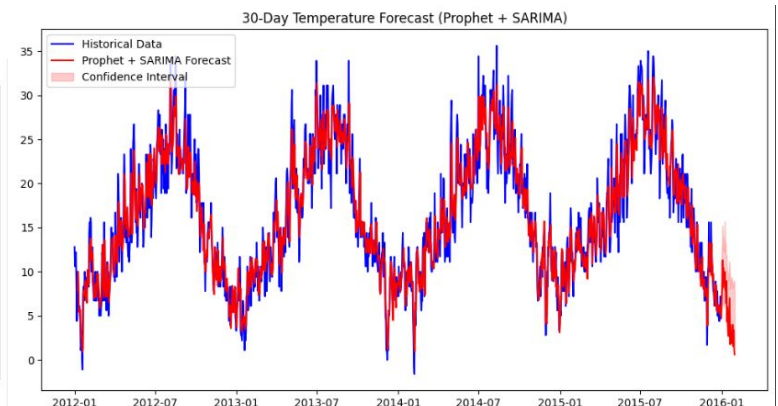
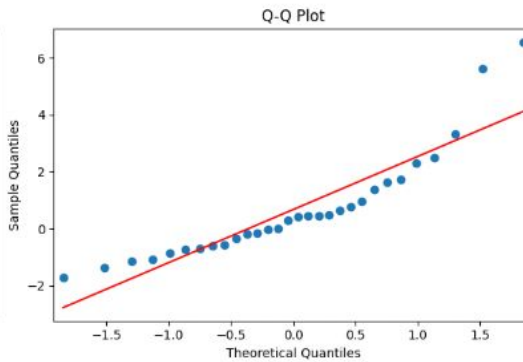
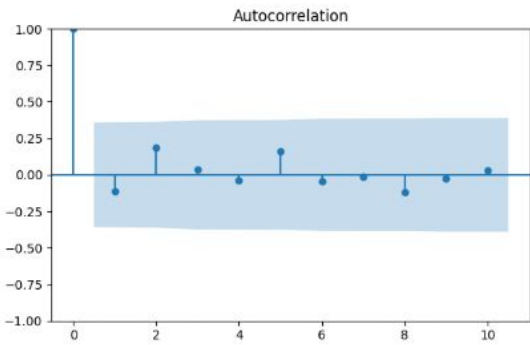
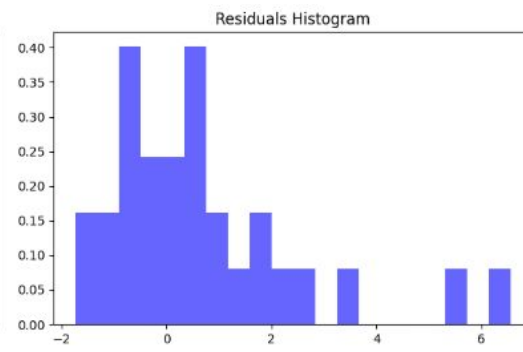
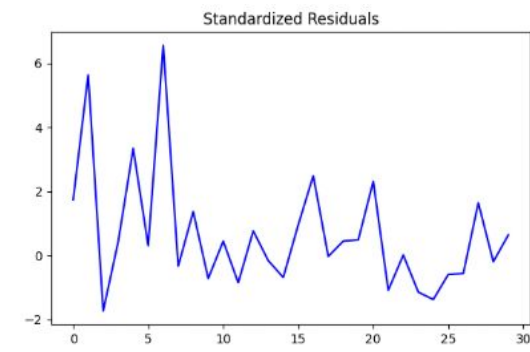
Q-Q Plot



Impulse Response of Prophet + Regressors



Prophet Corrected SARIMA



6.Final Conclusions

Best model: The combination of Prophet + SARIMA showed the best results among the tested models.

Key findings:

- SARIMA handles seasonality well but does not account for external factors.
- Prophet allows for the inclusion of additional variables but requires residual correction.
- Prophet + SARIMA combines the strengths of both approaches, improving forecast accuracy.

Model quality evaluation:

- Prophet + SARIMA had the lowest errors (MAE, RMSE).
- Residual analysis confirmed that SARIMA handles the temporal structure of the data better.